



# CAUSAL NETWORKS

Eg.



①  $\text{fire} \rightarrow \text{Smoke}$

$P(\text{Fire}, \text{Smoke})$     ?  $P(\text{Fire})$   
                                  $P(\text{Smoke}|\text{fire})$

②  $\text{Smoke} \rightarrow \text{Fire}$

$P(\text{Fire}, \text{Smoke})$     ?  $P(\text{Smoke})$   
                                  $P(\text{Fire}|\text{Smoke})$

# Key points

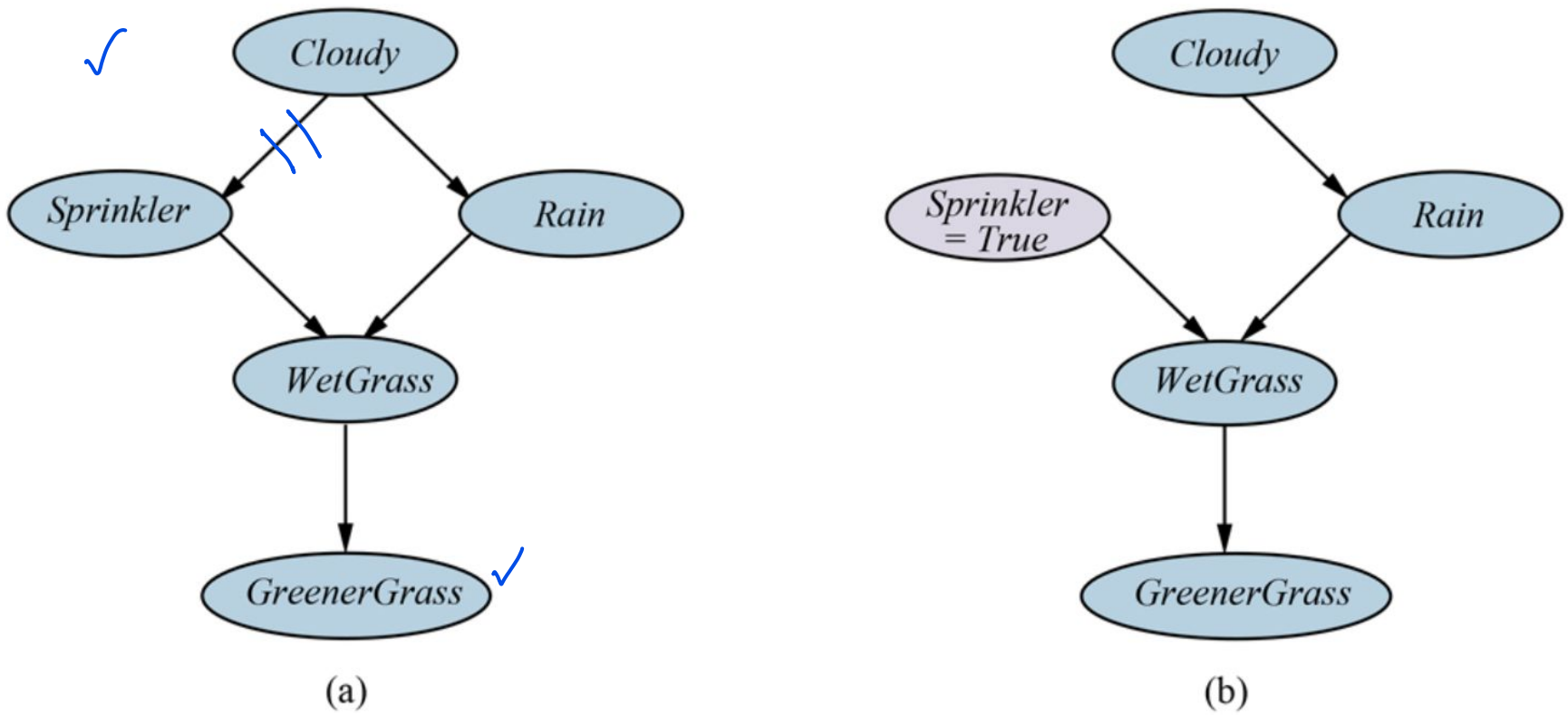
$$\underline{A} \rightarrow B$$

→ Causal Bayesian networks, sometimes called Causal Diagrams, were devised to permit us to represent causal asymmetries and to leverage the asymmetries towards reasoning with causal information. The idea is to decide on arrow directionality by considerations that go beyond probabilistic dependence and invoke a totally different type of judgment

\* In other words, the value  $x_i$  of each variable  $X_i$  is determined by an equation  $x_i = f_i(\text{OtherVariables})$ , and an arrow  $X_j \rightarrow X_i$  is drawn if and only if  $X_j$  is one of the arguments of  $f_i$ .

The equation  $x_i = f_i(\cdot)$  is called a **structural equation**, because it describes a stable mechanism in nature which, unlike the probabilities that quantify a Bayesian network, remains invariant to measurements and local changes in the environment.

Figure 13.23



(a) A causal Bayesian network representing cause-effect relations among five variables. (b) The network after performing the action "turn Sprinkler on."

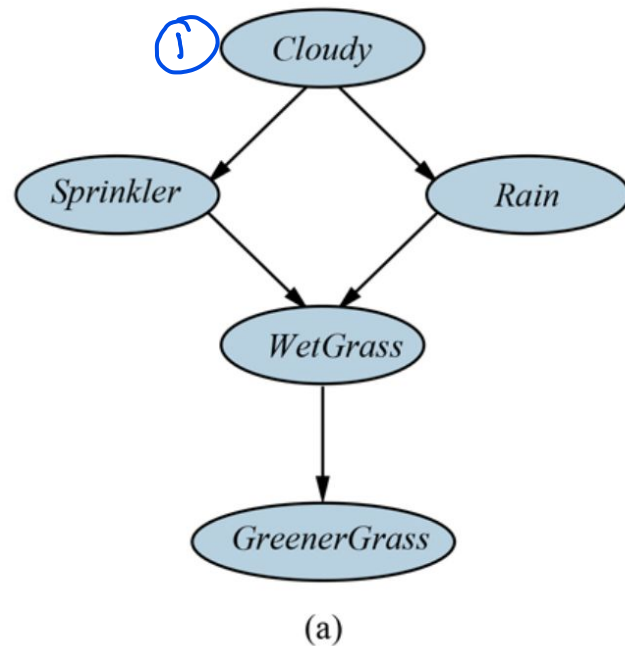
## Joint Probability

$$P(c,r,s,w,g) = P(c) P(r|c) P(s|c) P(w|r,s) P(g|w) \quad \text{--- ①}$$

## Structural Equations

$$\begin{aligned} \text{① } C &= f_C(U_C) \\ \text{② } R &= f_R(C, U_R) \\ \text{③ } S &= f_S(C, U_S) \\ \text{④ } W &= f_W(R, S, U_W) \\ \text{⑤ } G &= f_G(W, U_G) \end{aligned}$$

The U-variables in these equations represent unmodeled variables, also called error terms or disturbances, that perturb the functional relationship between each variable and its parents. For example,  $U_W$  may represent another potential source of wetness, in addition to Sprinkler and Rain— perhaps MorningDew or FirefightingHelicopter.



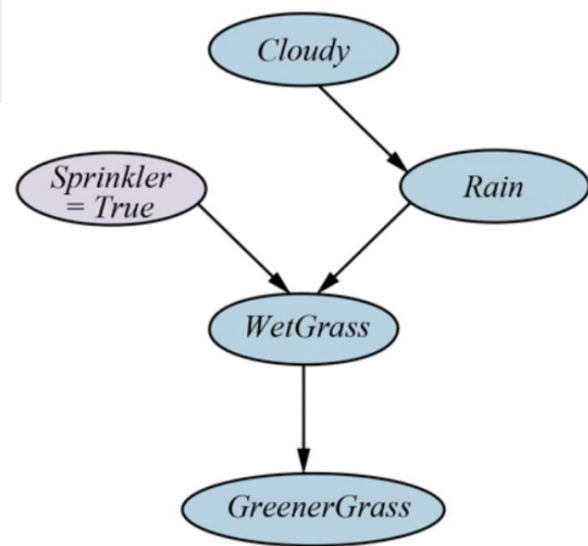
$$C = f_C(U_C) \quad \text{--- ①}$$

$$R = f_R(C, U_R) \quad \text{--- ②}$$

$$S = \text{true} \quad \text{--- ③}$$

$$W = f_W(R, S, U_W) \quad \text{--- ④}$$

$$G = f_G(W, U_G). \quad \text{--- ⑤}$$



$$P(c, r, w, g | \underbrace{do(S = \text{true})}) = P(c) P(r|c) P(w|r, s = \text{true}) P(g|w)$$



②

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i)). \quad \text{--- (I)}$$

After applying  $\underbrace{do(X_j = x_{jk})}$ , the new joint distribution  $P_{x_{jk}}$  simply omits the factor for  $X_j$ :

$$P_{x_{jk}}(x_1, \dots, x_n) = \begin{cases} \prod_{i \neq j} P(x_i | \text{parents}(X_i)) = \frac{P(x_1, \dots, x_n)}{P(x_j | \text{parents}(X_j))} & \text{if } x_j = x_{jk} \\ 0 & \text{if } x_j \neq x_{jk} \end{cases} \Rightarrow \text{(II)}$$

③

$$\begin{aligned} P(X_i = x_i | \underbrace{do(X_j = x_{jk})}) &= P_{x_{jk}}(X_i = x_i) \\ &= \sum_{\text{parents}(X_j)} \underbrace{P(x_i | x_{jk}, \text{parents}(X_j)) P(\text{parents}(X_j))}. \end{aligned}$$

The probability terms in the sum are obtained by computation on the original network, by any of the standard inference algorithms. This equation is known as an **adjustment formula**; it is a probability-weighted average of the influence of  $X_j$  and its parents on  $X_i$ , where the weights are the priors on the parent values. The effects of intervening on multiple

# Backdoor Criterion

$$\textcircled{1} S = \text{true} \rightarrow \text{Wet} \rightarrow \underline{GrGr}$$

$$\textcircled{\times} \textcircled{2} \text{Rain} \rightarrow \text{Wet} \rightarrow GrGr$$

- we might know that she checks the weather before deciding whether to turn on the sprinkler, but we might not know how she makes her decision
- The specific reason this is problematic in this instance is that we would like to predict the effect of turning on the sprinkler on a downstream variable such as GreenerGrass, but the adjustment formula must take into account not only the direct route from Sprinkler, but also the “back door” route via Cloudy and Rain.
- If we knew the value of Rain, this back-door path would be blocked—which suggests that there might be a way to write an adjustment formula that conditions on Rain instead of Cloudy

$$\Rightarrow \underline{P(g|do(S = true))} = \sum_r \underline{P(g|S = true, r)} \underline{P(r)}$$